Aaditya Sakhardande

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EDUCATION

Arizona State University, Tempe, AZ, USA — ***May 2026***
Masters in Robotics and Autonomous Systems (Electrical Engineering) — 3.7 GPA

Mukesh Patel School of Technology Management and Engineering, Mumbai, India — ***August 2024***
B.Tech. Major in Electronics and Telecommunication — 3.11 GPA

WORK EXPERIENCE

GrayMatter Robotics, Los Angeles, CA – *Robotics Engineer* — ***January 2026 – May 2026***
- Programmed and deployed FANUC robots for automated scanning and sanding of complex production parts, including on-site system activation and integration at customer facilities.
- Integrated LMI, Zivid, Alvium, and Cognex vision systems to enable high-precision 3D scanning and adaptive robot motion for irregular geometries.
- Developed ROS2-based robotic applications and managed Docker systems for scalable, reliable deployment across development and production environments.
- Extended robotic sanding workflows with error detection mechanisms for over-sanding and under-sanding, integrating feedback for improved precision on complex geometries.

Leyline Technologies, Spokane, WA - *Product Engineer* — ***June 2025 – August 2025***
- Tasked to design and document 3 AI models that optimize product features in alignment with real-time robotics systems.
- Iterating 2 data-driven prototypes, leveraging Python and internal tools to improve model accuracy and pipeline clarity by 20%.
- Collaboration with cross-functional teams to integrate product requirements into modular designs for robot-adjacent use cases.

Cartken | Arizona State University, Tempe, AZ – *Prototype Team Lead* — ***May 2025 – August 2025***
- Designed and prototyped robotic subsystems using reclaimed components, focusing on hardware and embedded control.
- Analyzed robotic platforms' environmental impact and hardware design, sustainable manufacturing, and reuse strategies.
- Applied electronic design and embedded programming to implement core functions in autonomous robotic systems.

MaitriAI, Mumbai, IN – *Software Developer* — ***December 2023 - May 2024***
- Trained CNN model reaching 98% accuracy for document identification. Enhanced Biometric OCR Software, extracting text with 94% precision from scanned documents. Implemented SQL database to store extracted documents and texts.
- Conceptualized enterprise architecture of LMS platform, enabling creation and management of training programs.
- Built and deployed 12 APIs employing Python libraries such as PyTesseract, TensorFlow, and FlaskAPI.

MPSTME University, Mumbai, IN – *Student Researcher* — ***August 2023 – November 2023***
- Led **Online Canteen Ordering Management System** project, building a web app for online orders, customizable menus, and tracking 300+ order statuses in real time.
- Leveraged and maintained system using **HTML, CSS, JavaScript, PHP, MySQL, and XAMPP server**.
- Coordinated UI design, integrated payment gateway, and optimized SQL schema for transactions and metadata storage.

RESEARCH PUBLICATIONS

Robotic Gait Trainer with Exoskeleton *(Link to the Paper)* — ***February 2026***
- Headed research on robotic gait trainer for spinal cord injury patients, conducting a 25-year trend analysis of functional models.
- Analyzed rehabilitation practices and BCI strategies to develop six optimal models for neural function recovery.
- Presented at the *International Conference on Materials, Robotics, Automation, Computer and Control (ICMRACC 2025)*.

PROJECTS

Drone Vision Navigation Framework *(Project)* — ***Spring 2025 – Summer 2025***
- Engineered a real-time vision system for the Parrot Mambo UAV drone using MATLAB/Simulink, achieving 92.3% landing accuracy across 50+ test flights via closed-loop feedback control.
- Used HSV color segmentation and centroid localization on a 640x480 video feed at 30 fps to extract spatial features in real-time.
- Designed control laws using normalized image-plane error, cutting lateral drift by over 83% and improving convergence speed.

Real-time Video Inpainting Software *(Project)* — ***Spring 2025***
- Built an object removal pipeline using Mask R-CNN and OpenCV (96% mask accuracy, 30 fps), integrating GAN-based inpainting models (LaMa, Stable Diffusion) for texture-consistent restoration.
- Optimized for achieved 1,000 frames per minute, supporting videos up to 1080p resolution with minimal artifacts.

Autonomous Maze Solver using MyCobot Pro 600 *(Project)* — ***Fall 2024 – Spring 2025***
- Engineered and built digital twin to simulate forward and inverse kinematics, ensuring precise and efficient navigation.
- Programmed MyCobot Pro 600 to autonomously solve a 4x4 maze, optimizing path planning and obstacle detection.
- Automated and executed a maze-solving algorithm incorporating Python, MATLAB, and SOLIDWORKS, merging socket programming and image processing for real-time maze recognition.

TECHNICAL SKILLS

Programming: Python, C++, JavaScript, Bash, Pandas, OpenCV, TensorFlow, PyTorch, SQL, PHP.
Software: MATLAB/Simulink, ROS2, Linux, Git, Anaconda, AutoCAD, SolidWorks, RepetierHost, Proteus, NerfStudio.
Other Skills: Project Management, GANs, Docker & REST APIs, Image Segmentation, PLCs, Control Systems (PID, Filter), NLP.
Certifications: 100 Days of Code (Python – Udemy), Advanced Python (LinkedIn), Learning C# (LinkedIn)